```
Start coding or generate with AI.
```

```
import sys
!{sys.executable} -m pip install transformers datasets torch pandas scikit-learn
print("Libraries installed successfully.")
```

```
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ment already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages
```

```
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ment already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from ma
es installed successfully.
```

```
from datasets import load_dataset

# Load the 'imdb' dataset
imdb_dataset = load_dataset('imdb')

# Print the dataset to inspect its structure
print(imdb_dataset)
print("Dataset loaded successfully.")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https:/
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public mode
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests t
```

README.md: 7.81k/? [00:00<00:00, 466kB/s]

plain_text/train-00000-of-00001.parquet: 100% 21.0M/21.0M [00:00<00:00, 105MB/s]

plain_text/test-00000-of-00001.parquet: 100% 20.5M/20.5M [00:00<00:00, 102MB/s]

plain_text/unsupervised-00000-of-00001.p(…): 100% 42.0M/42.0M [00:00<00:00, 109MB/s]

Generating train split: 100% 25000/25000 [00:00<00:00, 102736.44 examples/s]

Generating test split: 100% 25000/25000 [00:00<00:00, 86400.38 examples/s]

Generating unsupervised split: 100% 50000/50000 [00:00<00:00, 120226.70 examples/s]

```
DatasetDict({
    train: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    test: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    unsupervised: Dataset({
        features: ['text', 'label'],
        num_rows: 50000
    })
})
Dataset loaded successfully.
```

```
from transformers import DistilBertTokenizerFast

# 1. Load the DistilBertTokenizerFast tokenizer
tokenizer = DistilBertTokenizerFast.from_pretrained('distilbert-base-uncased')

# 2. Define a tokenization function
def tokenize_function(examples):
    return tokenizer(examples['text'], truncation=True, padding=True)

# 3. Apply this tokenization function to all splits of the imdb_dataset
tokenized_datasets = imdb_dataset.map(tokenize_function, batched=True, remove_columns=

print("Dataset tokenization complete.")
print(tokenized_datasets)
```

```
tokenizer_config.json: 100%    48.0/48.0 [00:00<00:00, 2.12kB/s]

vocab.txt:    232k/? [00:00<00:00, 5.01MB/s]

tokenizer.json:    466k/? [00:00<00:00, 9.29MB/s]

Map: 100%    25000/25000 [00:32<00:00, 818.01 examples/s]

Map: 100%    25000/25000 [00:27<00:00, 879.99 examples/s]

Map: 100%    50000/50000 [01:02<00:00, 830.14 examples/s]

Dataset tokenization complete.
DatasetDict({
    train: Dataset({
        features: ['label', 'input_ids', 'attention_mask'],
        num_rows: 25000
    })
    test: Dataset({
        features: ['label', 'input_ids', 'attention_mask'],
        num_rows: 25000
    })
    unsupervised: Dataset({
        features: ['label', 'input_ids', 'attention_mask'],
        num_rows: 50000
    })
})
```

```
import torch

# 1. Extract the 'train' split
train_dataset = tokenized_datasets['train']

# 2. Extract the 'test' split (which is usually referred to as evaluation set in Hugg
eval_dataset = tokenized_datasets['test']

# 3. Print the structure of both datasets
print("Train Dataset Structure:", train_dataset)
print("\nEvaluation Dataset Structure:", eval_dataset)
print("Datasets prepared successfully.")
```

```
Train Dataset Structure: Dataset({
    features: ['label', 'input_ids', 'attention_mask'],
    num_rows: 25000
})

Evaluation Dataset Structure: Dataset({
    features: ['label', 'input_ids', 'attention_mask'],
    num_rows: 25000
})
Datasets prepared successfully.
```

```
from transformers import DataCollatorWithPadding
from torch.utils.data import DataLoader

# 1. Set the format of both train_dataset and eval_dataset to 'torch'
train_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask', 'label
```

```
eval_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask', 'label'

# 2. Instantiate DataCollatorWithPadding
data_collator = DataCollatorWithPadding(tokenizer=tokenizer)

# 3. Create DataLoader instances
batch_size = 8 # Adjust based on GPU memory
train_dataloader = DataLoader(train_dataset, shuffle=True, batch_size=batch_size, col
eval_dataloader = DataLoader(eval_dataset, batch_size=batch_size, collate_fn=data_col

print("PyTorch Datasets and DataLoaders created successfully.")
print(f"Train DataLoader batch size: {train_dataloader.batch_size}")
print(f"Evaluation DataLoader batch size: {eval_dataloader.batch_size}")
```

```
PyTorch Datasets and DataLoaders created successfully.
Train DataLoader batch size: 8
Evaluation DataLoader batch size: 8
```

```
from transformers import DistilBertForSequenceClassification

# 1. Load the pre-trained DistilBertForSequenceClassification model
# 2. Specify num_labels=2 for binary sentiment classification (positive/negative)
model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased'

print("Pre-trained DistilBertForSequenceClassification model loaded successfully.")
# 3. Optionally, print the model architecture to verify
print(model)
```

config.json: 100% 483/483 [00:00<00:00, 38.0kB/s]

model.safetensors: 100% 268M/268M [00:01<00:00, 231MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 412.07it/s, Materializing param=distilbert.transformer.layer.5.sa

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.weight | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.weight       | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok
- MISSING   :those params were newly initialized because missing from the checkpoint. (
Pre-trained DistilBertForSequenceClassification model loaded successfully.
DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
          (attention): DistilBertSelfAttention(
            (q_lin): Linear(in_features=768, out_features=768, bias=True)
            (k_lin): Linear(in_features=768, out_features=768, bias=True)
            (v_lin): Linear(in_features=768, out_features=768, bias=True)
            (out_lin): Linear(in_features=768, out_features=768, bias=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
            (lin1): Linear(in_features=768, out_features=3072, bias=True)
            (lin2): Linear(in_features=3072, out_features=768, bias=True)
            (activation): GELUActivation()
          )
          (output_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
        )
      )
    )
  )
  (pre_classifier): Linear(in_features=768, out_features=768, bias=True)
  (classifier): Linear(in_features=768, out_features=2, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
)
```

```
from transformers import DistilBertForSequenceClassification

# 1. Load the pre-trained DistilBertForSequenceClassification model
# 2. Specify num_labels=2 for binary sentiment classification (positive/negative)
model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased'

print("Pre-trained DistilBertForSequenceClassification model loaded successfully.")
# 3. Optionally, print the model architecture to verify
print(model)
```

```
Loading weights: 100%   100/100 [00:00<00:00, 426.49it/s, Materializing param=distilbert.transformer.layer.5.sa

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
pre_classifier.weight    | MISSING    |
pre_classifier.bias      | MISSING    |
classifier.weight        | MISSING    |
classifier.bias          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok
- MISSING   :those params were newly initialized because missing from the checkpoint. (
Pre-trained DistilBertForSequenceClassification model loaded successfully.
DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
          (attention): DistilBertSelfAttention(
            (q_lin): Linear(in_features=768, out_features=768, bias=True)
            (k_lin): Linear(in_features=768, out_features=768, bias=True)
            (v_lin): Linear(in_features=768, out_features=768, bias=True)
            (out_lin): Linear(in_features=768, out_features=768, bias=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
            (lin1): Linear(in_features=768, out_features=3072, bias=True)
            (lin2): Linear(in_features=3072, out_features=768, bias=True)
            (activation): GELUActivation()
          )
          (output_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
        )
      )
    )
  )
  (pre_classifier): Linear(in_features=768, out_features=768, bias=True)
  (classifier): Linear(in_features=768, out_features=2, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
)
```

```
from transformers import TrainingArguments

# 2. Instantiate TrainingArguments
```

```
training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=3,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    warmup_steps=500,
    weight_decay=0.01,
    logging_dir='./logs',
    logging_steps=10,
    # Temporarily removing evaluation_strategy and save_strategy due to TypeError,
    # and disabling load_best_model_at_end to resolve the subsequent ValueError.
    # Ideally, these would be set to 'epoch' and match.
    load_best_model_at_end=False, # Changed to False to bypass ValueError
    metric_for_best_model='accuracy',
    report_to='none'
)

print("Training Arguments defined successfully.")
# 3. Print the instantiated TrainingArguments object to verify the settings
print(training_args)
```

```
save_on_each_node=False,
save_only_model=False,
save_steps=500,
save_strategy=SaveStrategy.STEPS,
save_total_limit=None,
seed=42,
skip_memory_metrics=True,
tf32=None,
torch_compile=False,
torch_compile_backend=None,
torch_compile_mode=None,
torch_empty_cache_steps=None,
trackio_space_id=trackio,
use_cache=False,
use_cpu=False,
use_liger_kernel=False,
warmup_ratio=None,
warmup_steps=500,
weight_decay=0.01,
)
```

```
import sys
!{sys.executable} -m pip install evaluate
print(" 'evaluate' library installed successfully.")
```

```
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: datasets>=2.0.0 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: dill in /usr/local/lib/python3.12/dist-packages (from ev
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: requests>=2.19.0 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: tqdm>=4.62.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: multiprocess in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fsspec>=2021.05.0 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: huggingface-hub>=0.7.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (fro
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packa
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/di
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packa
```

```
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from h
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (fro
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (f
Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84.1/84.1 kB 5.1 MB/s eta 0:00:00
Installing collected packages: evaluate
Successfully installed evaluate-0.4.6
 'evaluate' library installed successfully.
```

```
import evaluate
import numpy as np
```